

Detailed Review of the Main Paper and Supplement

Risk Bounds for Quantile Temporal-Spatial Analysis

Technical, Mathematical, and Editorial Comments

30 July 2026

Page-number convention. Every page reference below refers to the supplied 133-page compiled PDF, not to the revised PDF.

Scope and overall assessment

The manuscript has a promising objective: risk bounds for conditional-quantile estimation with temporal dependence, a shared spatial random effect, heavy-tailed measurement error, and several structured function classes. The current compiled version, however, contains foundational mathematical inconsistencies in the dependence assumptions, coupling, localization, and application-specific calculations. It also contains literal internal review notes in the compiled appendix. These issues are more serious than ordinary copyediting.

The companion full-length redline preserves the paper’s section order—general theory, trend filtering, high-dimensional linear quantile regression, K -NN fused lasso, nonconvex classes, neural networks, algorithms, experiments, and the complete proof supplement—in one TeX file. In the main paper, mathematically invalid or internally inconsistent statements are replaced by corrected model, assumption, theorem, and algorithm blocks. Appendices A–I now contain operative corrected statements and proofs in red; the previous appendix comments have been replaced by actual derivations. A final gray tracked-comparison appendix retains the supplied detailed derivations solely so that every change can be located against the original source. The experimental numerical values and designs are not altered; experiment-specific concerns are listed separately below.

Regenerated complete version. The unified redline contains the full main paper, operative corrected appendices, the tracked source comparison, and the references in one file. Red text gives the implemented revisions. Gray text in the final comparison appendix reproduces the supplied derivations only for auditability and is not part of the operative proof. No appendix comment is left as an unresolved instruction.

Most important substantive changes in the revised paper:

1. beta mixing is imposed on the *joint time cluster* $W_i = (X_i, \delta_i, E_i)$, Berbee coupling is applied to that joint object, and the design law used after conditioning is made explicit;
2. the empirical processes are centered at f^* and localized through an explicit critical radius;
3. the tail remainder is quantitatively tied to the curvature margin;
4. prediction and parameter-recovery claims are separated in the high-dimensional linear section;
5. graph-TV scaling and normalized loss conventions are corrected;
6. the nonconvex result uses a genuine near-best approximating function and a centered star hull;
7. the neural-network class has a specified, clipped, sparse architecture; and
8. the ADMM dual step is corrected.

Scope of the mathematical repair. The internal logical gaps identified in the appendix have been addressed: the proof now uses joint-cluster coupling, centered weighted processes, consistent conditioning, explicit graph and network conditions, and a fixed-point radius. Source-dependent ingredients are isolated as named assumptions rather than asserted without support. The resulting document is a mathematically coherent revision, but—as with any new theoretical argument—the authors should still check cited external approximation, entropy, and graph-geometry inputs against the primary sources before submission.

Priority legend

Critical means the issue invalidates a theorem/proof step or makes the stated claim undefined. **Major** means the claim is materially ambiguous, inconsistent, or overstated. **Minor** means the issue is localized but should still be corrected.

A. Mathematical and theoretical issues

ID	PDF page(s)	Priority	Location	Issue	Recommended correction	Revision status
T01	2, 7	Critical	Model and Assumption 1.1	The data-generating model is never given a dedicated equation. Assumption 1.1 says the data are generated from “model (1),” but Equation (1) is the estimator, not a stochastic model.	State explicitly that $y_{ij} = f^*(x_{ij}) + \delta_i(x_{ij}) + e_{ij}$ and separately impose the conditional-quantile calibration. Refer to this equation throughout.	Rewritten
T02	2	Critical	Notation: empirical inner product	The displayed empirical inner product contains $f(x)g(x) dx$ inside a discrete sum, and the population integrals/suprema use $[0, 1]$ even though the domain is $[0, 1]^d$.	Define $\langle f, g \rangle_{n,m} = n^{-1} \sum_i m_i^{-1} \sum_j f(x_{ij})g(x_{ij})$ and integrate over $[0, 1]^d$ with respect to the design law Q .	Corrected
T03	2	Major	Conditional quantile definition	The equality $P\{y_{ij} \leq f^*(x_{ij}) \mid x_{ij}\} = \tau$ presumes an exact, effectively unique conditional quantile. Quantiles may be set-valued when the conditional CDF has a jump.	Define $f^*(x) = \inf\{t : F_{ij}(t \mid x) \geq \tau\}$, then impose the exact calibration/continuity condition only when it is needed for Knight’s identity to cancel the linear term.	Clarified
T04	3, 32, 125-127	Critical	Blocking construction	The sets I_s are written as two-point sets such as $\{(s-1)L, sL\}$ instead of integer intervals, and the modular indexing is ambiguous at zero and at the remainder block.	Use interlaced sets $\mathcal{J}_{e,\ell} = \{\ell + 2qL_n : q \geq 0\} \cap [n]$ and $\mathcal{J}_{o,\ell} = \{\ell + (2q+1)L_n : q \geq 0\} \cap [n]$. These sets partition $[n]$ and have a clear separation gap.	Rewritten
T05	3, 11-12	Critical	Theorem 1 statement	The theorem uses a localization radius t that is not defined or quantified. The displayed O_P statement therefore does not specify a rate or a fixed-point condition.	Define a blockwise localized complexity $\Psi_n(r)$ and a critical radius $r_n = \inf\{r : \Psi_n(r) \leq cr^2\}$. State the risk as $O_P(r_n^2)$.	Rewritten
T06	3-4, 11-12	Major	Generic “polylog” rate	The text claims a generic $\text{polylog}(n)/(nm)$ rate for broad classes. Such a rate cannot follow without entropy, Gaussian-width, or other geometric control of the particular class.	Keep the main result at the localized-complexity level. Evaluate the fixed point only in the trend-filtering, linear, graph, and network applications.	Corrected
T07	4, 18, 95	Critical	Projection in the nonconvex case	$f_{\text{Proj}} := \inf_{f \in \mathcal{F}} \ f - f^*\ _\infty$ defines a scalar, but subsequent arguments treat it as a function.	Choose an approximate minimizer $f^\circ \in \mathcal{F}$ satisfying $\ f^\circ - f^*\ _\infty \leq \inf_{f \in \mathcal{F}} \ f - f^*\ _\infty + n^{-1}$ and define ϕ_n from that function.	Corrected
T08	17-18	Major	Star hull definition	The definition writes “for any $\mu \in [0, 1]$ ” outside the set and does not cleanly distinguish the affine star hull of functions from the centered star hull of errors.	Use $\mathcal{H}_{\mathcal{F}}^* = \{\mu(f - f^*) : f \in \mathcal{F}, \mu \in [0, 1]\}$ and localize this centered class in $L_2(Q)$.	Corrected

Continued on next page

ID	PDF page(s)	Priority	Location	Issue	Recommended correction	Revision status
T09	7	Critical	Independence versus beta mixing	Assumption 1 describes $\{\delta_i\}$ and $\{e_{ij}\}$ as independently distributed and then imposes nontrivial beta mixing across time. Read literally, the assumptions contradict one another.	Assume conditional independence within a time cluster and beta mixing across time for the joint cluster process. Remove claims of cross-time independence.	Rewritten
T10	7, 32-35, 125-127	Critical	Separate couplings	The proof couples X_i , E_i , and δ_i separately. Three unrelated couplings need not preserve their contemporaneous joint law, so equalities involving y_{ij} in the “star world” do not follow.	Apply Berbee’s lemma once to $W_i = (X_i, \delta_i, E_i)$. The coupled cluster then has the correct joint marginal law and is independent within each interlaced block.	Rewritten
T11	7-8, 42-43, 52	Critical	Within-cluster dependence and design law	Several proofs symmetrize over all (i, j) as if the summands were conditionally independent. The mean-remainder calculation also integrates against the marginal design law Q after conditioning on δ_i , which requires independence of the sampling locations and the spatial effect.	State that X_i is independent of δ_i , that the x_{ij} are conditionally i.i.d. from Q , and that the e_{ij} are independent conditional on (δ_i, X_i) ; otherwise use cluster-level arguments and the correct conditional design law.	Assumption strengthened
T12	8	Critical	Assumption 2	The condition is stated for “any any function ϕ ” and uses an absolute-value CDF inequality without clearly specifying the scalar perturbation, the conditioning point, or uniformity over (i, j) .	State a uniform scalar local-curvature condition: for $ u \leq R$, $\underline{f} u \leq F_{ij}(f^*(x) + u x) - F_{ij}(f^*(x) x) \leq \bar{f} u $.	Rewritten
T13	8-10	Major	Assumption 3: definition of Π	The conditioning alternates between x , x_S , and the full vector, and it is unclear whether the same active set S works for every block and every i .	Define $D_i(x, u)$ first, then its block average $\Pi_I(x, u)$. State explicitly that dependence on x is only through one fixed coordinate set S and that the bound holds uniformly over all blocks.	Rewritten
T14	8-10, 53-54	Critical	Dimension of the grid argument	The uniform grid covers s design coordinates and the threshold u , but the union bound uses M^s rather than M^{s+1} . The threshold dimension is missing.	Use M^{s+1} grid points and carry the resulting $(s + 1)$ factor into the logarithmic term.	Corrected
T15	10, 49-55	Critical	Tail constant and curvature absorption	The small tail constant $\tilde{c}_0$ is not quantitatively related to the curvature margin. Later proofs redefine the penalty term in an attempt to absorb it.	Fix $c_0 < C_0$ once and require the tail constant to satisfy an explicit inequality such as $8C_m\eta_0 < C_0 - c_0$. Never redefine the curvature penalty later.	Rewritten
T16	10-11, 55-56	Major	Tail example	The definition of T_n is repeated/circular, and conditional tail probabilities are bounded by unconditional high-probability events without carefully conditioning on x and δ_i .	Present the tail assumption as a primitive condition. Give a separate sufficient example with explicit independence and uniform envelope conditions, or remove the informal derivation.	Consolidated

Continued on next page

ID	PDF page(s)	Priority	Location	Issue	Recommended correction	Revision status
T17	11-12, 41-43	Critical	Uncentered Rademacher class	The complexity is taken over f rather than $f - f^*$. Consequently the class need not contain zero, the supremum can be negative, and the proof cannot simply drop the coupling indicator by using $1_\Omega \leq 1$.	Use the centered class $\{f - f^* : f \in \mathcal{F}\}$ in every Rademacher process. The class then contains zero and the supremum is nonnegative.	Corrected
T18	11-12	Major	Block normalization	The theorem sums normalized block complexities without giving a fixed-point equation that accounts for $2L$ blocks and their effective sizes. Statements later insert or remove factors of L inconsistently.	Define the block process with the original empirical weights $n^{-1}m_i^{-1}$, or equivalently multiply the within-block normalized complexity by its block weight $M_{a,\ell}/N$, before summing the blocks.	Corrected in unified revision
T19	12, 30, 60	Major	Huber-type loss and L_2 conversion	The manuscript alternates between empirical and population Huber losses and asserts an L_2 comparison with factor B . Since both f and f^* are bounded by B , their difference may be as large as $2B$.	Use a consistent population loss $\Delta^2(h) = \int \min\{ h , h^2\} dQ$ and the bound $\ h\ _2^2 \leq (1 \vee 2B)\Delta^2(h)$.	Corrected
T20	12-14	Critical	Trend-filtering domain	Total variation and weak derivatives are written on $[0, 1]^d$, while the application and spline arguments are univariate. The formula $\int_{[0,1]^d} f^{(k)}(t) dt$ is not a valid univariate derivative identity for $d > 1$.	State explicitly that the trend-filtering application has $d = 1$ and define $J_k(f) = \text{TV}(f^{(k-1)})$ on $[0, 1]$.	Corrected
T21	12	Major	Polynomial degree/indexing	The text says $J_k(f) = \text{TV}(f^{(k-1)})$ but then claims the minimizer is a degree- k spline; the examples correspond instead to degree $k - 1$. It also cites Equation (2), which is the block definition.	Use one convention throughout: $k = 1$ piecewise constant, $k = 2$ piecewise linear, and degree $k - 1$. Cite the estimator equation, not the block equation.	Corrected
T22	13	Critical	Radius $V = 1$	Canonical scaling $V^* = O(1)$ does not imply $V = 1$ or $J_k(f) \leq 1$. Rescaling the constraint changes the class and the constants/rate.	Keep a general radius $V \geq J_k(f^*)$ and show its exponent in the rate. Set $V = 1$ only after an explicit normalization.	Corrected
T23	13	Major	Duplicated theorem setup	Theorem 3 repeats the estimator and assumptions twice, once with a generic seminorm and once with total variation, and alternates between f and $\hat{f}$.	State one estimator, one class, and one theorem. Define all symbols before the display.	Corrected
T24	13-14, 63-66	Critical	Trend-filtering exponents	The powers of B and $\log n$ differ between Theorem 3, Lemma 39, and the proof conclusion. Several algebraic steps used to solve the fixed point are not valid.	Recompute the entropy integral and solve one explicit fixed point. The revised statement gives the standard $N^{-2k/(2k+1)}$ exponent, a displayed V dependence, and a separate spatial-effect term.	Rewritten
T25	14	Major	Minimax claim	The text calls the rate minimax optimal without requiring the additional spatial term to be no larger than the nonparametric term. If that term dominates, the displayed upper bound does not match the cited lower bound.	Qualify the statement: the usual minimax exponent is recovered only when the spatial-effect remainder is asymptotically no larger than the trend-filtering term.	Corrected

Continued on next page

ID	PDF page(s)	Priority	Location	Issue	Recommended correction	Revision status
T26	14	Major	Linear model constraint	Assumption 6.2 writes $V \geq V^* \geq \ \beta\ _1$ rather than $\ \beta^*\ _1$, and the treatment of an intercept is not specified.	Write $f^*(x) = x^\top \beta^*$ with $\ \beta^*\ _1 \leq V$. Include an intercept in x if desired and state how the bounded-predictor condition is enforced.	Corrected
T27	15	Major	Design-matrix definition	The dimensions and row definition of X are malformed, and A is introduced without a clear block normalization. This makes the theorem difficult to check dimensionally.	Define the weighted block matrix with row $x_{ij}^\top/m_i$ and use $n^{-1} \sum_{a,\ell} E \max_q \ X_{a,\ell,q}^w\ _2$, which matches the estimator's weights.	Corrected in unified revision
T28	5-6, 15	Critical	Coefficient-consistency claim	The prose claims consistent coefficient estimation and variable selection without a restricted-eigenvalue or identifiability condition. The displayed risk controls only prediction.	State the result as a prediction bound. Explicitly say that coefficient error and support recovery still require a restricted-eigenvalue or comparable condition.	Corrected
T29	15	Major	Linear-rate summary	The expression " $(nm)^{-1/2} + n^{-1}s$ " is ambiguous and does not agree with the theorem's dependence on V , $\log d$, block sizes, or the spatial term.	Derive the simplified rate only after column normalization and retain V , $\sqrt{\log d}$, and all block/log factors.	Corrected
T30	16	Critical	Graph total-variation scaling	Equation (21) states $\ \nabla_{G_K} \theta^*\ _1 \asymp (nm)^{-1/d}$. An unnormalized sum of edge differences cannot decay at that rate; the expected scaling is approximately $N^{1-1/d}$ up to logarithms.	Replace the negative exponent by $N^{1-1/d} \text{polylog}(N)$ and propagate that scaling into the theorem.	Corrected
T31	15-16	Major	Symmetric K -NN graph	The prose says points are connected if one is a nearest neighbor of the other "and vice versa," which describes the mutual graph, whereas the intended construction elsewhere is the symmetrized graph using "or."	State explicitly whether the graph is mutual or symmetric. The revised paper uses the symmetric union graph.	Clarified
T32	17, 68-75	Critical	KNN norm scaling	Theorem 5 writes an unnormalized Euclidean norm on an N -vector but gives a rate of order $N^{-1/d}$. The dimensions do not match unless the norm is normalized.	Use $\ u\ _{n,m}^2 = n^{-1} \sum_i m_i^{-1} \sum_j u_{ij}^2$ (or $N^{-1} \ u\ _2^2$ when $m_i \asymp m$).	Corrected
T33	17	Major	Undefined graph quantities	V_{G^*} is not defined, and the explanation changes $\text{polylog}(nm)$ into $\text{poly}(nm)$, which would destroy consistency.	Use $V_n \geq \ D_{G_K} \theta^*\ _1$ and write $\text{polylog}(N)$ consistently.	Corrected
T34	17	Minor	Extension rule	The denominator in the out-of-sample K -nearest-neighbor average is written as an indicator sum without stating $K \leq N$ or addressing ties.	Define a deterministic tie-breaking rule, assume $K \leq N$, and write the average over an explicitly defined set $N_K(x)$ of size K .	Clarified
T35	17-18	Critical	Nonconvex theorem contradiction	Theorem 7 is titled as a nonconvex result but says "let $\mathcal{F} \dots$ be convex." This contradicts both the section and the proof strategy.	Remove the convexity assumption and use the star-hull argument based on convexity of the loss, not of the class.	Corrected

Continued on next page

ID	PDF page(s)	Priority	Location	Issue	Recommended correction	Revision status
T36	18	Major	Approximation error notation	The text writes $\phi_{nm} := \sup_{f_{\text{Proj}} \in \mathcal{F}} \ f_{\text{Proj}} - f^*\ _\infty$, which is the worst approximation rather than the best approximation.	Define ϕ_n from a best or near-best approximant. Do not take a supremum over the class.	Corrected
T37	18, 95–103	Critical	Two-sided empirical process	The nonconvex comparison introduces both signs of the empirical process, but subsequent corollaries copy one-sided convex bounds without consistently reversing every decomposition.	State a symmetric localized process bound on the centered star hull and use it for both signs. Avoid duplicating twelve nearly identical terms.	Rewritten
T38	19–20	Critical	Neural-network recursion	The displayed recursion applies the ReLU inside $f_i^{(s)}$ and then applies another ReLU to $f_i^{(L-1)}$ in the output. Layer and coefficient indices are also inconsistent.	Define $h_0 = x$, $h_\ell = \sigma(W_\ell h_{\ell-1} + b_\ell)$ once per hidden layer, followed by an affine output.	Corrected
T39	20–21	Critical	Network class not controlled	The class specifies only depth and width. It has no weight, sparsity, or output constraint, so the global boundedness assumption and the stated entropy/risk bound do not follow.	Use a clipped-output network class and specify sparsity/parameter count (or weight bounds). Evaluate complexity for that exact class.	Rewritten
T40	20–21	Major	Hierarchical composition assumptions	The set $\mathcal{P}$, number of levels, number of nodes, and relevant bounded domains are not required to be finite/fixed, although the approximation theorem depends on these quantities.	Assume a finite bottleneck set and a fixed finite hierarchy. State boundedness of intermediate functions/domains.	Strengthened
T41	21	Critical	Unspecified architecture in Theorem 10	The rate is asserted for arbitrary L and r , but no choice balances approximation and estimation. Such a rate cannot hold uniformly over all architectures.	Make L_N , width, and sparsity part of the theorem and select them to balance approximation error and localized complexity.	Rewritten
T42	22	Major	Trend-filtering penalty order	The algorithm section alternates among $TV^{(k)}(f)$, $D^{(k+1)}$, and the theory's $TV(f^{(k-1)})$. The convention is not consistent.	Adopt one indexing convention and define the discrete operator to match it. The revised algorithm uses the same $D^{(k)}$ convention as its class definition.	Corrected
T43	22–24	Critical	ADMM dual step	The “dual update” is written as an argmax of a positive quadratic, which is unbounded and is not an ADMM update.	Use the scaled-dual update $u^{t+1} = u^t + \theta^{t+1} - z^{t+1}$ and monitor primal and dual residuals.	Corrected
T44	23	Major	Check-loss proximal formula	The case analysis is phrased in terms of $\theta^{(t-1)}$ even though the optimization variable is θ_{ij} . This obscures the proximal map and its thresholds.	Define $q = z^t - u^t$ and write the closed-form proximal update directly in three cases according to the location of y_{ij} relative to q .	Rewritten
T45	24–25	Major	KNN ADMM statement	The dual update maximizes over z instead of u , the text contains “is is,” and Algorithm 2 is captioned as trend filtering rather than graph fused lasso.	Correct the optimization variable, caption, and update equations.	Corrected

Continued on next page

ID	PDF page(s)	Priority	Location	Issue	Recommended correction	Revision status
T46	30–31	Major	Population and empirical loss notation	Δ_{nm}^2 and Δ^2 use inconsistent measures and the notation $P(t)$ is not defined. Several later displays silently switch between them.	Reserve $\Delta_{n,m}^2$ for the weighted empirical average and define Δ^2 as an integral with respect to Q .	Corrected
T47	31–32	Major	Excess-loss definitions	Some displays subtract $\rho_\tau(y_{ij} - f^*(x_{ij}))$, while a later line substitutes the same term inconsistently as $\rho_\tau(y_{ij} - f^*)$ instead of $\rho_\tau(\delta_i + e_{ij})$.	Define one excess loss $\widehat{Z}_{ij}(f)$ and use the model to rewrite its baseline exactly once.	Consolidated
T48	32	Minor	Population minimizer wording	The text says $f^* \in \arg \min Z(f)$ even though Z is already defined as excess risk. The intended facts are $Z(f^*) = 0$ and $Z(f) \geq 0$.	State the excess-risk properties directly.	Corrected
T49	34	Major	Coupling-error probability	Lemma 16 replaces $3/n$ by $\epsilon/2$ without stating the required lower bound on n and then uses the bound as if it were uniform.	Keep the explicit $3/n$ term until the final asymptotic step, or state “for $n \geq 6/\epsilon$.”	Corrected
T50	34–35	Critical	Independence inside a block	The proof says the entire collection indexed by (i, j) is i.i.d. within a block. Berbee coupling only makes the time clusters independent; within-cluster observations may remain dependent through δ_i .	Say clusters are independent across i . Use conditional independence within i only after conditioning on δ_i , or use cluster-level symmetrization.	Corrected
T51	36–37	Major	Curvature outside the local neighborhood	The lower-bound proof uses the Huber threshold one even though the local CDF condition is valid on $[-R, R]$. Constants and thresholds therefore require rescaling by R .	Derive $C_0 \min\{ h , h^2\}$ with C_0 depending explicitly on R and $\underline{f}$, or define a Huber function with threshold R .	Corrected in constants
T52	37	Minor	Corollary 22 annotation	The source itself contains the note “the notation here does not match the statement from (68),” confirming that the population expectation is not consistently defined.	Define $\overline{Z}(f) = E\widehat{Z}(f)$ and $\Delta^2 = E\Delta_{n,m}^2$ before the corollary.	Corrected
T53	41–43	Critical	Literal internal comments in lemmas	Reviewer/author notes such as “this does not make sense,” “same comment from before,” and “please check everywhere” are typeset as part of the mathematical statements.	Remove all internal notes. Replace the malformed class by $\{f \in \mathcal{F} : c_0\Delta^2(f - f^*) \leq t^2\}$ and center the process.	Removed
T54	42–43	Critical	Conditional symmetrization	The proof calls $\widehat{Z}_{ij}^* - \widetilde{Z}_{ij}^*$ independent over all (i, j) after conditioning, but that conclusion requires a precise within-cluster conditional-independence assumption.	Add that assumption explicitly and condition on the full cluster effect/design; otherwise symmetrize independent cluster sums.	Rewritten
T55	44	Major	Lemma 27 conditioning	A literal note says the right-hand side needs conditioning on x . The lemma’s statement and proof use inconsistent conditional probabilities.	Condition every probability on (x_{ij}, δ_i) in $\widetilde{Z}_{ij}$ and on x_{ij} in Z_{ij} .	Corrected

Continued on next page

ID	PDF page(s)	Priority	Location	Issue	Recommended correction	Revision status
T56	45–48	Major	Remainder algebra	The derivation of $Z_{ij}(f) - \tilde{Z}_{ij}(f)$ contains a parenthesis error, repeated terms, and the note “first equality seems wrong.” The final cancellation is not presented transparently.	Apply Knight’s identity once to both conditional expectations; subtract them and cancel the baseline terms in one aligned display.	Rewritten
T57	49, 51–55	Critical	Penalty $\ell(f)$ redefined	$\ell(f)$ is first $(C_0 - c_0)\Delta^2$ and is later set to unrelated multiples of $\tilde{c}_0\Delta^2$. This breaks the localization argument.	Choose c_0 and $\ell(f)$ once. Absorb tail terms by an assumption that makes their coefficient smaller than $C_0 - c_0$.	Corrected
T58	49	Major	Incomplete proof text	Corollary 30 contains “by Lemma 29 incomplete sentence” and “Oscar pauses here,” so the proof is not publication-ready.	Delete internal notes and provide a complete explanation of each inequality.	Removed
T59	50–51	Critical	Ghost design sample with shared random function	The ghost construction changes x_{ij} while sharing δ_i and e_{ij} , then asserts i.i.d. symmetry. Since $\delta_i(x)$ is evaluated at the design, this requires a carefully specified conditional kernel and is not automatic.	Formulate the cluster as one random element and apply standard conditional symmetrization to independent clusters. Avoid ad hoc partial ghost samples.	Rewritten
T60	52–55	Critical	Average versus sum of absolute values	The grid argument controls the absolute value of a block average of signed remainders, but Step 3 replaces a sum of absolute values by the absolute value of the average. That inequality is false in general.	Assume/control the signed block average that actually appears in the empirical-process decomposition, without inserting absolute values term by term.	Rewritten
T61	53–54	Critical	Hoeffding centering	Hoeffding’s inequality is applied to the block average Π without explicitly showing that the summands are independent and centered at each grid point. Heterogeneous cluster laws can invalidate centering.	Either assume common cluster marginals within each block or define $D_i - ED_i$ and track the deterministic mean separately.	Assumption made explicit
T62	54	Major	Incorrect localization indicators	The proof partitions with $1\{f - f^* \leq 1\}$ and $1\{f - f^* > 1\}$, omitting absolute values and therefore mishandling negative deviations.	Use $1\{ f - f^* \leq 1\}$ and its complement.	Corrected
T63	55–56	Major	Tail-example conditioning	The example treats events such as $\ \delta_i\ _\infty \leq R_n$ as if they directly bound probabilities conditional on the realized δ_i . The conditioning and high-probability layers are mixed.	Separate the outer high-probability envelope event from the inner conditional tail bound and state all independence assumptions.	Rewritten with explicit localization range
T64	57–58	Critical	A_3 ghost-process argument	Z_{ij} is already a conditional expectation depending on the design, but the ghost construction includes δ^* and e^* and asserts a symmetry that is not justified as written.	Treat $Z_{ij}(f)$ as a deterministic function of the independent coupled cluster/design and apply standard symmetrization at that level.	Rewritten
T65	58–60	Critical	Circular/corrupted aggregation	The proof of Lemma 34 says “Thus, by Theorem 1” while it is still proving Theorem 1, repeatedly substitutes $J_{e,1}$ for $J_{e,\ell}$, and drops the coupling remainder.	Aggregate the block lemmas directly, retain the sum over ℓ , and add the explicit coupling probability before invoking the critical-radius lemma.	Rewritten

Continued on next page

ID	PDF page(s)	Priority	Location	Issue	Recommended correction	Revision status
T66	60–63	Critical	Radius t versus t^2	Several local balls use radius proportional to t^2 where the L_2 radius should be proportional to t . The final fixed-point display therefore has inconsistent dimensions.	If the Huber excess is at most t^2 , use $\ f - f^*\ _2 \lesssim t$. Keep squared risks and radii distinct.	Corrected
T67	61–62	Major	O_P criterion	The displayed limit in Corollary 38 mixes κ , r_n , $3/n$, and the complexity terms in a malformed fraction, and the proof says an inequality holds “and the last” without a complete argument.	Use a standard peeling/fixed-point lemma with a monotone sub-root complexity and state the high-probability result explicitly.	Rewritten
T68	63–66	Critical	Trend-filtering population-to-empirical localization	The proof invokes a Bartlett et al. corollary to assert a set inclusion between population and empirical balls with a probability expression that is dimensionally and notationally incorrect.	Use a valid entropy bound uniformly over the empirical design metric, or first prove a separate norm-equivalence event for the bounded TV class.	Rewritten
T69	65	Critical	Fixed-point algebra	The implication involving powers of b and N is algebraically false, and later displays interchange B , L , and the block length. This is the source of conflicting exponents.	Redo the entropy integral symbolically and solve the scalar inequality once; do not absorb parameter powers by unsupported comparisons.	Rewritten
T70	66	Major	Lemma/theorem mismatch	Lemma 39 and the proof of Theorem 3 give different powers of B and $\log n$. The theorem cannot be verified from the lemma.	Use one rate throughout and show the final multiplication by the Huber-to- L_2 factor explicitly.	Corrected
T71	67–68	Major	Linear proof denominator	The odd-block bound divides by $\sum_{i \in J_{e,\ell}} m_i$ rather than the odd-block size. The proof also inserts an additional L without clearly tracing normalization.	Use $M_{a,\ell}$ for each block and keep the block normalization in the definition of $\mathcal{A}_n$.	Corrected
T72	71–72	Critical	KNN Rademacher supremum	The initial complexity display has $s_{ij}\theta_{ij}^*$ under a supremum over θ , so the supremum no longer depends on the candidate. The subsequent centering step does not follow from that display.	Start with $\sum s_{ij}(\theta_{ij} - \theta_{ij}^*)$ over the centered graph-TV class.	Corrected
T73	72–75	Critical	Circular local-radius assumption	The KNN proof assumes $b^2 = O(n^{-1} + N^{-1})$ in order to discard the local-radius term, although b is the unknown radius the proof is meant to determine.	Keep the b term and solve the resulting fixed-point inequality; do not assume the desired rate.	Rewritten
T74	73–75	Major	Logarithmic exponents and symbols	The proof alternates among l , ℓ , and μ in powers such as $\log^{3+l}(N)$ and gives inconsistent choices of K .	Choose one $K = \text{polylog}(N)$ and use a named, fixed logarithmic exponent depending only on d .	Consolidated
T75	78–82	Critical	KNN extension proof	The proof contains an unresolved “(??)” reference, swaps the labels Int_1 and Int_2 , and uses dimensionally incorrect radii such as $(1/2 + 1/N) + R_{K,\max}$.	Reorganize the extension error into estimation and approximation components, use radii of order $1/N + R_{K,\max}$, and provide valid references.	Rewritten

Continued on next page

ID	PDF page(s)	Priority	Location	Issue	Recommended correction	Revision status
T76	83-93	Major	Dependent-design KNN geometry	Several occupancy and nearest-neighbor lemmas are copied from an i.i.d. setting. The block coupling is invoked, but the probability bounds and union factors do not consistently reflect $2L_n$ blocks and N^d cells.	State a single graph-regularity proposition under the joint mixing design and prove/quote the required occupancy and degree events with all block and cell factors.	Consolidated
T77	93-94	Critical	Graph lemmas: equality/norm errors	Lemma 49 writes an equality where only a triangle-inequality bound is available, and Lemma 50 alternates between D^+ and $D^\dagger$ and between ℓ_1 and ℓ_2 graph-TV norms.	Use inequalities throughout, one pseudoinverse notation, and the ℓ_1 incidence norm that defines graph total variation.	Corrected
T78	95-98	Critical	Nonconvex localization typos	The proof assumes the infimum projection is attained, substitutes an undefined f , and later contains $\widehat{Z}(f) - \widehat{Z}(f)$ where a population/empirical difference was intended.	Use a near-minimizer f° , compare $\widehat{Z}(\widehat{f})$ with $\widehat{Z}(f^\circ)$, and write both empirical-process signs correctly.	Rewritten
T79	95-100	Major	$\widehat{t}^2$ definitions	The approximation-adjusted radius is defined with different factors in Equations (214), (215), and the following display, so the event and threshold do not match.	Define one radius, for example $\widehat{t}^2 = t^2 + C\phi_n^2$, and use it unchanged.	Corrected
T80	99-103	Critical	Sign errors in twelve-term decomposition	The decomposition of $\overline{Z} - \widehat{Z}$ and its reverse interchanges $Z - \widetilde{Z}$ with $\widetilde{Z} - Z$ in several places. Therefore the named A , E , and F terms do not equal the preceding process.	Avoid the twelve-term duplication. Bound the absolute centered process on the star hull once.	Rewritten
T81	104-105	Major	Wrong block in copied corollary	The bound for $\widehat{B}_1$ uses $\mathcal{R}_{J_{e,\ell}}$ instead of the odd-block complexity.	Use the matching odd block and verify every copied even/odd label.	Corrected by consolidation
T82	108-113	Critical	Missing weights and inconsistent expectations	Some E_1 displays omit $m_i^{-1} \sum_j$, and expectations unnecessarily include Rademacher variables before symmetrization. The coefficient of $\ell(f)$ changes again.	Retain the weighted cluster average in every term and use one conditional-expectation hierarchy and one penalty coefficient.	Rewritten
T83	114-124	Major	Network entropy/application proof	The network application does not clearly connect the approximation theorem, pseudo-dimension/entropy bound, block sample sizes, and the localized fixed point. The architecture parameters remain implicit.	State a lemma with explicit approximation and entropy bounds for the clipped sparse class, then balance approximation and estimation in the theorem.	Rewritten
T84	125-127	Critical	Berbee lemma proof	The coupling proof invokes "Theorem 5" rather than the stated mixing theorem and its induction assumes that each new coupled variable is independent of all earlier coupled variables, which is precisely what must be constructed.	Quote sequential Berbee coupling for the joint clusters or prove it by conditioning on the sigma-field generated by all previously coupled clusters.	Rewritten

Continued on next page

ID	PDF page(s)	Priority	Location	Issue	Recommended correction	Revision status
T85	128-133	Major	Reference list accuracy	The references contain duplicate authors, incomplete publication data, broken capitalization, PDF-extraction hyphenation, and entries that appear to have swapped meta-data.	Regenerate the bibliography from a verified Bib-Tex database and check every cited theorem against the correct edition/page/theorem number.	Cleaned; meta-data check remains

B. Grammar, wording, and presentation issues

The table lists representative errors. Many constructions recur, so the entire source should still receive a final professional copyedit after the mathematics is settled.

ID	PDF page(s)	Location	Issue	Recommended edit
G01	1	Abstract	“a broad prototyped function class” is not idiomatic and does not describe the mathematical object.	Use “a broad class of candidate functions.”
G02	1	Abstract	“The function class is convex or non-convex” is a fragment and the following “For convex case” lacks an article.	Write two complete sentences distinguishing the convex and nonconvex results.
G03	1	Abstract	“pioneering advancement” is promotional and unsupported.	State precisely which results are new.
G04	1	Abstract	The abstract says the paper uses “simulated and real datasets,” but the supplied manuscript presents simulations and no identifiable real-data analysis.	Remove the real-data claim unless a real-data section is added.
G05	1–2	Introduction	“inclusivity towards error distributions” and similar phrases are vague.	Use “applicability to heavy-tailed error distributions.”
G06	2	Model paragraph	The sentence beginning “Then given a quantile level” has multiple misplaced modifiers and lacks a main verb.	Split the definition of f^* from the statement of the estimation goal.
G07	2	Notation	“Given a distribution Q ..., and $\{z_{ij}\}$ identically distributed” is a sentence fragment.	Write “Let Q be ... and let $z_{ij} \sim Q$.”
G08	3	Summary	“let $f^*(x_{ij})\tau$ -be conditional quantiles” is ungrammatical.	Write “let $f^*(x_{ij})$ be the conditional τ -quantile.”
G09	3	Summary	“without any moments condition requirements” is wordy and ungrammatical.	Use “without moment assumptions.”
G10	3	Blocking paragraph	“Denote for” and “if L has the order” are incorrect constructions.	Use “Define” and “if L is of order $\log n$.”
G11	4	Convex/nonconvex summary	“release the condition” should be “relax the condition.”	Replace and rewrite the surrounding sentence.
G12	4	Nonconvex summary	“boarder class” is a spelling error.	Use “broader class.”
G13	4	Rate discussion	“The A_1 equals to” is ungrammatical.	Use “The term A_1 is ...”.
G14	4	Boundedness paragraph	“to target a sharp convergence rates” has article/number disagreement.	Use “to obtain sharp convergence rates.”
G15	4–5	Proof overview	Several sentences exceed 50 words and repeat “convert” four times.	Separate the loss comparison, symmetrization, and localization steps into distinct paragraphs.
G16	5	Trend-filtering discussion	“1th derivative” should be “first derivative” or “ $(k - 1)$ st derivative,” depending on context.	Correct ordinal notation throughout.
G17	6	Literature review	“Theory for deep networks includes ... and so on” is a run-on list with missing spaces before citations.	Replace with a focused paragraph on approximation and statistical-risk results relevant to this paper.

Continued on next page

ID	PDF page(s)	Location	Issue	Recommended edit
G18	6	Neural-network paragraph	"the estimator models neural network achieves" is ungrammatical.	Use "the neural-network estimator achieves."
G19	6, 21	Terminology	"Quantial" appears instead of "Quantile."	Correct the spelling throughout.
G20	6-7	General methods	"We let the class of estimators possesses" has subject-verb disagreement.	Use "We assume that the function class satisfies ...".
G21	6	Convexity	"which signifying that" should be "meaning that."	Rewrite the sentence.
G22	7	Nonconvexity	"the later case" should be "the latter case."	Correct the word.
G23	8	Assumption 2	"for any any function" repeats a word.	Delete the duplicate and use a scalar perturbation.
G24	8	Assumption 3	"That is, indicates with k -th entry" is not grammatical.	Write "That is, x^k is obtained by replacing the k th coordinate ...".
G25	8	Assumption 3	Missing spaces produce strings such as " u)for."	Repair spacing and run a LaTeX-aware spell check.
G26	9-10	Assumptions discussion	"Theorem 1 replies on" appears twice.	Use "Theorem 1 relies on."
G27	10	Example 2	"If we have ... satisfied, Then" is a comma splice with incorrect capitalization.	Split into two sentences.
G28	11	Example 2	"condition condition" repeats a word.	Delete the duplicate.
G29	11-12	Theorem discussion	"this addition term" should be "this additional term."	Correct the phrase.
G30	12	Rate discussion	Write $\log n$ consistently rather than "logn."	Standardize mathematical typography.
G31	13	Theorem 3	The setup is duplicated verbatim and includes "And let."	Delete the duplicate and avoid beginning formal statements with "And."
G32	15	Theorem 4 discussion	"normalizeda standard assumption" is missing punctuation/spacing.	Write "normalized, a standard assumption ...".
G33	15-16	KNN introduction	"has showed" should be "has shown."	Correct the verb.
G34	16	Graph discussion	"the x_{ij} s" is awkward.	Use "the design points x_{ij} ."
G35	16	Assumption 7	"Under Assumptions 7" should be singular.	Use "Under Assumption 7."
G36	18	Nonconvex discussion	"guranntees" is misspelled.	Use "guarantees."
G37	19	Neural-network introduction	"Our papar" is misspelled.	Use "Our paper."
G38	21	Network theorem	"an hierarchical" should be "a hierarchical."	Correct the article.
G39	21	Rate comparison	"is the same to the rate" should be "is the same as the rate."	Correct the preposition.
G40	24	Algorithm discussion	"is is" and "(51)can" are spacing/duplication errors.	Correct and proofread the paragraph.
G41	25	Simulation design	"vary n in in" and "For each fix value" are errors.	Use "vary n over ..." and "for each fixed value."
G42	26	Signal description	"Sinoidal" and "constructed to have with non-homogenous smoothness" are incorrect.	Use "Doppler-type function with spatially inhomogeneous smoothness."
G43	27	Figure 2 caption	The caption says "Panels a, b, and b."	Use "Panels a, b, and c."
G44	28-29	Nonconvex experiment	"i.i.d.-style pairs" is misleading because flattening does not make dependent panel observations i.i.d.	Use "flattened input-response pairs" without implying independence.

Continued on next page

ID	PDF page(s)	Location	Issue	Recommended edit
G45	30	Acknowledgments	The JMLR template instruction remains as manuscript text.	Replace it with the actual funding and competing-interest statements.
G46	41–49	Proof appendix	Internal directives to coauthors/reviewers appear in the compiled paper.	Remove all comments from production text; keep them only as TeX comments or a separate review file.
G47	50, 58, 107, 110	Proof appendix	“Combines (99) and (100)” appears repeatedly.	Use “Combining (99) and (100) ...”.
G48	50, 106, 109	Proof appendix	“differed on” is used where “differ only in” is intended.	Rewrite the ghost-sample descriptions.
G49	51, 107, 110	Proof appendix	“on the both side” is ungrammatical.	Use “on both sides.”
G50	70	KNN appendix	“The rest poofs” contains two errors.	Use “The remaining proofs.”
G51	81	KNN extension	“It reminds to analyze” should be “It remains to analyze.”	Correct the sentence.
G52	109	Nonconvex appendix	“seperately” is misspelled.	Use “separately.”
G53	Throughout	Capitalization	Terms such as “Trend Filtering,” “High-dimensional Quantile Linear Regression,” and “Empirical Process Theory” are capitalized inconsistently.	Use sentence case except for proper names and section titles.
G54	Throughout	Terminology	The manuscript alternates among “temporal-spatial,” “spatial-temporal,” and “spatio-temporal.”	Choose one term and use it consistently; the revision uses “temporal-spatial” in the title and “spatio-temporal” only where conventional.
G55	Throughout	Hyphenation and spacing	PDF extraction artifacts such as “sub-Gaussian,” broken words, and missing spaces occur frequently.	Compile from clean source, remove soft-hyphen artifacts, and run automated spell/grammar checks.

C. Experiment-specific concerns (flagged, not substantively changed)

Per the requested scope, these items were not repaired in the revised paper beyond local grammar and LaTeX cleanup. They should be resolved before submission because some affect reproducibility or whether the simulated target is the asserted conditional quantile.

ID	PDF page(s)	Location	Concern	Suggested action	Status
E01	25, 28	Basis function definition	Equation (53) appears to define $h_t(x)$ as a 25-dimensional vector, whereas $\delta_i(x)$ needs a scalar and Equation (59) later uses a product over coordinates.	Confirm the implemented scalar basis and make Equations (53) and (59) identical.	Not changed
E02	25–26	Design recursion with unequal m_i	When $m_i > m_{i-1}$, the rule $x_{ij} = x_{i-1,j}$ is not defined for newly appearing indices.	Specify a padded array of size $M = \max_i m_i$ before truncation, as is done later in the neural-network study.	Not changed
E03	25–26	Quantile target under correlated noise	The text says a normalization stabilizes variance when $\tau \neq 0.5$, but variance normalization does not by itself make the conditional τ -quantile of $\delta_i(x) + e_{ij}$ equal zero.	Describe the centering/quantile shift used to enforce the target quantile in every scenario.	Not changed
E04	26	BIC sample size and weights	The BIC formula uses n observations and un-weighted residuals even though the estimator uses $N = \sum_i m_i$ observations with cluster weights $1/(nm_i)$.	Explain which effective sample size and weighted loss were implemented.	Not changed
E05	26	Degrees of freedom	The degrees-of-freedom estimate uses a threshold of 0.05 without justification and the notation “ $d + k + 1$ ” conflicts with dimension d .	Rename the knot count and provide a citation/sensitivity check for the threshold.	Not changed
E06	28–29	Flattening dependent data	Flattening creates a matrix representation but does not create i.i.d. training pairs.	Avoid “i.i.d.-style” wording and state how minibatches, validation splits, and standard errors respect cluster/time dependence.	Grammar changed only
E07	28–29	Scenario numbering	The multivariate section lists Scenarios 1–4 and the network section jumps to Scenarios 6–8; Scenario 5 is absent.	Renumber or explain the omitted scenario.	Not changed
E08	29–30	Scenario 8 target	Scenario 8 declares $\tau = 0.5$ but evaluates a grid $q \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$. The oracle quantile for each q is not fully specified under the dynamic noise process.	Give the oracle q -quantile function used in the MSE calculation for every q .	Not changed
E09	29–30	Monte Carlo uncertainty	Table 1 reports only means, with no standard errors, confidence intervals, or number of trials stated next to the table.	Report Monte Carlo standard errors and the exact number of replications for each row.	Not changed
E10	29–30	Neural-network comparability	Architecture, optimizer, learning rate, epochs, early stopping, random seeds, and hyperparameter selection are not reported.	Add a reproducibility table and confirm identical tuning budgets for SqErr and Quantile networks.	Not changed

ID	PDF page(s)	Location	Concern	Suggested action	Status
E11	29-30	Interpretation of squared-error baseline	A mean-regression network is not designed to estimate tail quantiles, so its poor $q = 0.1, 0.9$ performance is expected and is not a strong quantile-regression benchmark.	Add competitive quantile-network, quantile-forest, and/or distributional-regression baselines.	Not changed
E12	30	Acknowledgments/funding	The manuscript contains only template instructions, not an actual disclosure.	Insert verified funding and conflict-of-interest statements.	Author action

D. Implementation map for the unified revision

Revised section	Main changes implemented
Abstract and Introduction	Replaced promotional and grammatically unstable claims with a precise model, target, estimator, and contribution summary. Removed the unsupported claim of a real-data study.
General Methods	Added the stochastic model, corrected empirical/population norms, imposed joint-cluster beta mixing, defined local quantile curvature, and introduced an explicit blockwise critical radius.
Trend Filtering	Restricted the argument to $d = 1$, corrected derivative/degree indexing, retained a general TV radius V , and stated a single internally consistent rate.
High-Dimensional Linear Regression	Recast the result as a prediction theorem, defined block design matrices clearly, and removed coefficient/support-recovery claims that require restricted eigenvalues.
K -NN Fused Lasso	Corrected graph-TV scaling, normalized the prediction norm, defined the symmetric graph and incidence matrix, and separated observed-point estimation from nearest-neighbor extension error.
Nonconvex Theory	Replaced the scalar “projection” with a near-best function, used the centered star hull, and bounded both signs of one localized empirical process.
Neural Networks	Corrected the layer recursion; specified clipped sparse ReLU classes and architecture sequences; connected approximation, entropy, and the nonconvex theorem.
Algorithms	Replaced the unbounded “dual argmax” with the scaled-dual update and wrote the check-loss proximal step explicitly.
Experiments	Preserved designs, numbers, and interpretations; corrected grammar, captions, and LaTeX only. Experiment-specific concerns remain listed in Section C of this review.

Appendices	Replaced the former proof comments with operative red statements and proofs for the convex theorem, trend filtering, linear prediction, graph fused lasso, nonconvex localization, neural networks, ADMM, auxiliary inequalities, and beta mixing. Joint-cluster coupling, full-objective weights, centered local classes, and the critical-radius argument are implemented directly. The supplied detailed appendix is retained in gray only as a tracked comparison, with red implemented-correction passages.
References	Converted the PDF reference list to a compilable in-file bibliography. Metadata should still be checked against a clean BibTeX database.

E. Recommended pre-submission checklist

1. Confirm that the strengthened joint-process and conditional-independence assumptions match the intended data-generating mechanism.
2. Confirm that the deliberately explicit logarithmic factors and architecture/graph assumptions match the preferred primary-source formulations.
3. Check every external theorem used for entropy, graph geometry, and neural-network approximation against the original source; the internal implications are now proved conditionally on those inputs.
4. Reconcile the two basis-function definitions and document how every simulated error distribution is shifted to have conditional τ -quantile zero.
5. Add full computational details, uncertainty measures, and competitive quantile baselines.
6. Replace the acknowledgment placeholder with verified funding and competing-interest disclosures.
7. Rebuild the references from verified BibTeX records and run a final grammar/style check after all mathematical revisions are accepted.

Bottom line

The supplied manuscript was not submission-ready because several core proof steps were undefined or invalid and internal comments remained in the compiled appendix. The

regenerated redline implements a coherent restart around the joint-cluster dependence structure, full-objective block weighting, centered localization, and an explicit fixed point. It replaces the appendix comments by actual red derivations and retains the original detailed source in gray for comparison. The experimental substance remains unchanged. Before submission, the authors should still validate the external graph-geometry, entropy, and neural-network approximation inputs against the cited primary sources and resolve the experiment-specific items in Section C.